

undetermined coefficients

Can be used to solve non-homogenous linear DE's with constant coefficients when the right hand side is:

- 1) Polynomial, x, x^2, x^3
- 2) exponential function, e^{ax}
- 3) trig functions, $\sin(Bx), \cos(Bx)$
- 4) any linear combos of the above

• You can only apply this method when you are dealing with these if you come across a problem with the right hand side is something else you need to apply another method

Steps

- 1) Solve for the homogenous equation by setting the problem equal to zero $\Rightarrow y_c$
- 2) make a guess for the particular solution, y_p using trial particular table $\Rightarrow y_p$
- 3) compare y_p with y_c and see if there is any repetition if there is add a x to the term that is being repeated
- 4) depending on the given equation take the derivative of y_p and plug into equation

5) create general solution: $y = y_c + y_p$

Table 4.4.1

Trial Particular Solutions

$g(x)$	Form of y_p
1. 1 (any constant)	A
2. $5x + 7$	$Ax + B$
3. $3x^2 - 2$	$Ax^2 + Bx + C$
4. $x^3 - x + 1$	$Ax^3 + Bx^2 + Cx + D$
5. $\sin 4x$	$A \cos 4x + B \sin 4x$
6. $\cos 4x$	$A \cos 4x + B \sin 4x$
7. e^{3x}	Ae^{3x}
8. $(9x - 2)e^{5x}$	$(Ax + B)e^{5x}$
9. $x^2 e^{3x}$	$(Ax^2 + Bx + C)e^{3x}$
10. $e^{3x} \sin 4x$	$Ae^{3x} \cos 4x + Be^{3x} \sin 4x$
11. $5x^2 \sin 4x$	$(Ax^2 + Bx + C) \cos 4x + (Ex^2 + Fx + G) \sin 4x$
12. $xe^{3x} \cos 4x$	$(Ax + B)e^{3x} \cos 4x + (Cx + D)e^{3x} \sin 4x$

$$y'' - 8y' + 16y = 28x + 4 \quad g(x)$$

to find y_c we will set the equation to zero:

$$y'' - 8y' + 16y = 0 \longrightarrow m^2 - 8m + 16 = 0 \longrightarrow (m-4)(m-4) = 0$$

• this is a case #2

$$y_c = C_1 e^{4x} + C_2 x e^{4x}$$

we will use the table of trial of Particular Solutions to get y_p
notice $28x + 4$ looks like $5x + 7$ so:

$$y_p = Ax + B$$

now we will compare y_p with y_c to see if there is repetition:

A and C_1, C_2 ?	No repetition	} - no repetition
B and C_1 ?	No repetition	
B and C_2 ?	No repetition	

take derivative and plug into original equation

$$y_p = Ax + B$$

$$(0) - 8(A) + 16(Ax + B) = 28x + 4$$

$$y_p' = A$$

$$-8A + 16Ax + 16B = 28x + 4$$

$$y_p'' = 0$$

$$16Ax + (-8A + 16B) = 28x + 4$$

equate the coefficient's

$$16A = 28, \quad -8A + 16B = 4$$

$$A = \frac{7}{4}, \quad -8\left(\frac{7}{4}\right) + 16B = 4$$

$$-14 + 16B = 4$$

$$16B = 18$$

$$B = \frac{9}{8}$$

Now we know what A and B are we can plug them into y_p

$$y_p = Ax + B$$

$$y_p = \frac{7}{4}x + \frac{9}{8}$$

Now use $y = y_c + y_p$ to get the general solution

$$y = C_1 e^{4x} + C_2 x e^{4x} + \frac{7}{4}x + \frac{9}{8}$$

$$y'' + 2y = -18x^2 e^{2x}$$

$$y'' + 2y = 0 \rightarrow m^2 + 2 = 0 \rightarrow m = \pm i\sqrt{2}$$

• this is a case #3

$$y_c = C_1 \cos(\sqrt{2}x) + C_2 \sin(\sqrt{2}x)$$

$$y_p = (Ax^2 + Bx + C) e^{2x}$$

A and C_1 and C_2 ? No repetition

B and C_1 and C_2 ? No repetition

$$f = Ax^2 + Bx + C$$

$$g = e^{2x}$$

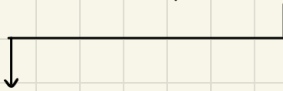
$$\frac{df}{dx} \cdot g + f \cdot \frac{dg}{dx}$$

$$\frac{df}{dx} = 2Ax + B$$

$$\frac{dg}{dx} = 2e^{2x}$$

$$y'_p = (2Ax + B) e^{2x} + (Ax^2 + Bx + C) 2e^{2x}$$

$$y'_p = (2Ax + B) e^{2x} + (Ax^2 + Bx + C) 2e^{2x}$$



$$f = (2Ax + B)$$

$$g = e^{2x}$$

$$\frac{df}{dx} = 2A$$

$$\frac{dg}{dx} = 2e^{2x}$$

$$\Rightarrow (2A)(e^{2x}) + (2Ax + B)(2e^{2x})$$



$$f = (Ax^2 + Bx + C)$$

$$g = 2e^{2x}$$

$$\frac{df}{dx} = 2Ax + B$$

$$\frac{dg}{dx} = 4e^{2x}$$

$$\Rightarrow (2Ax + B)(2e^{2x}) + (Ax^2 + Bx + C)4e^{2x}$$

$$y' = (2Ax + B)e^{2x} + (Ax^2 + Bx + C)2e^{2x}$$

$$y' = 2Ax e^{2x} + B e^{2x} + 2Ax^2 e^{2x} + 2Bx e^{2x} + 2C e^{2x}$$

$$\frac{2Ax e^{2x}}{f'g + f \cdot g'}$$

$$\Rightarrow (2A)(e^{2x}) + (2Ax)(2e^{2x})$$

$$\frac{2Ax^2 e^{2x}}{f'g + f \cdot g'}$$

$$\Rightarrow (4Ax)(e^{2x}) + (2Ax^2)(2e^{2x})$$

$$\frac{2Bx e^{2x}}{f'g + f \cdot g'}$$

$$\Rightarrow (2B)(e^{2x}) + (2Bx)(2e^{2x})$$

$$y'' = \cancel{2Ae^{2x}} + \cancel{4Ax e^{2x}} + \cancel{2Be^{2x}} + \cancel{4Ax^2 e^{2x}} + \cancel{4Ax^2 e^{2x}} + \cancel{28e^{2x}} + \cancel{4Bx e^{2x}} + 4C e^{2x}$$

$$y'' = (4Ax^2 e^{2x} + (4Ax e^{2x} + 4Ax e^{2x} + 4Bx e^{2x}) + (2Ae^{2x} + 2Be^{2x} + 2Be^{2x} + 4C e^{2x}))$$

$$y'' = (e^{2x} (4Ax^2 + (8Ax + 4Bx) + (2A + 4B + 4C)))$$

$$y_p = (Ax^2 + Bx + C) e^{2x}$$

$$(\cancel{e^{2x}} (4Ax^2 + (8Ax + 4Bx) + (2A + 4B + 4C))) + 2((Ax^2 + Bx + C) \cancel{e^{2x}}) = -18x^2 \cancel{e^{2x}}$$

$$\underline{4Ax^2} + \underline{8Ax} + \underline{4Bx} + \underline{2A} + \underline{4B} + \underline{4C} + \underline{2Ax^2} + \underline{2Bx} + \underline{2C} = -18x^2$$

$$x^2(4A + 2A) + x(8A + 4B + 2B) + (2A + 4B + 4C + 2C) = -18x^2$$

$$\cancel{x^2}(4A + 2A) = -18\cancel{x^2}, \quad x(8A + 4B + 2B) = 0, \quad (2A + 4B + 4C + 2C) = 0$$

$$6A = -18$$

$$\boxed{A = -3}$$

$$8A + 6B = 0$$

$$8(-3) = -6B$$

$$-24 = -6B$$

$$\boxed{B = 4}$$

$$2A + 4B + 4C + 2C = 0$$

$$2(-3) + 4(4) + 6C = 0$$

$$-6 + 16 = -6C$$

$$10 = -6C$$

$$\boxed{C = -\frac{5}{3}}$$

write a general solution: $y = y_c + y_p$

$$y = c_1 \cos(\sqrt{2}x) + c_2 \sin(\sqrt{2}x) + \left(-3x^2 + 4x - \frac{5}{3}\right)e^{2x}$$

$$y'' - y' + \frac{1}{4}y = 5 + e^{x/2}$$

$$y'' - y' + \frac{1}{4}y = 0 \quad \rightarrow \quad m^2 - m + 1/4 \quad \rightarrow \quad \text{apply quadratic formula} \quad \rightarrow \quad \begin{matrix} a=1 \\ b=-1 \\ c=1/4 \end{matrix}$$

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \Rightarrow \frac{-(-1) \pm \sqrt{(-1)^2 - 4(1)(1/4)}}{2(1)} = \frac{1 \pm 0}{2} \Rightarrow \frac{1}{2} \text{ repeated}$$

$$y_c = c_1 e^{x/2} + c_2 x e^{x/2}$$

$$y_p = A + B e^{x/2}$$

A and c_1, c_2 ? No repetition

B and c_1 ? yes, $y_p = A + B x e^{x/2}$

B and c_2 ? yes, $y_p = A + B x^2 e^{x/2}$

$$y_p = A + B x^2 e^{x/2}$$

← this will be the new y_p

$$f = Bx^2$$

$$g = e^{x/2}$$

$$\frac{df}{dx} \cdot g + f \cdot \frac{dg}{dx}$$

$$\frac{df}{dx} = 2Bx$$

$$\frac{dg}{dx} = \frac{1}{2} e^{x/2}$$

$$y'_p = (2Bx)(e^{x/2}) + (Bx^2)\left(\frac{1}{2}e^{x/2}\right)$$

$$y'_p = (2Bx)(e^{x/2}) + (Bx^2)(\frac{1}{2}e^{x/2})$$

$$y'_p = 2Bxe^{x/2} + \frac{1}{2}Bx^2e^{x/2}$$

$$\frac{2Bx}{f} \frac{e^{x/2}}{g}$$

$$f' \cdot g + f \cdot g'$$

$$(2B)(e^{x/2}) + (2Bx)(\frac{1}{2}e^{x/2})$$

$$\frac{\frac{1}{2}Bx^2}{f} \frac{e^{x/2}}{g}$$

$$f' \cdot g + f \cdot g'$$

$$(Bx)(e^{x/2}) + (\frac{1}{2}Bx^2)(\frac{1}{2}e^{x/2})$$

$$y''_p = 2Be^{x/2} + Bxe^{x/2} + Bxe^{x/2} + \frac{1}{4}Bx^2e^{x/2}$$

equations

$$y''_p = 2Be^{x/2} + 2Bxe^{x/2} + \frac{1}{4}Bx^2e^{x/2}$$

$$y'_p = 2Bxe^{x/2} + \frac{1}{2}Bx^2e^{x/2}$$

$$y_p = A + Bx^2e^{x/2}$$

$$(2Be^{x/2} + 2Bxe^{x/2} + \frac{1}{4}Bx^2e^{x/2}) - (2Bxe^{x/2} + \frac{1}{2}Bx^2e^{x/2}) + \frac{1}{4}(A + Bx^2e^{x/2}) = 5 + e^{x/2}$$

$$2Be^{x/2} + 2Bxe^{x/2} + \frac{1}{4}Bx^2e^{x/2} - 2Bxe^{x/2} - \frac{1}{2}Bx^2e^{x/2} + \frac{1}{4}A + \frac{1}{4}Bx^2e^{x/2} = 5 + e^{x/2}$$

$$2Be^{x/2} + \frac{1}{4}A = 5 + e^{x/2}$$

equate coefficients

$$2Be^{x/2} = e^{x/2}, \quad \frac{1}{4}A = 5$$

$$B = \frac{1}{2}$$

$$A = 20$$

write a general solution: $y = y_c + y_p$

$$y = c_1 e^{x/2} + c_2 x e^{x/2} + 20 + \frac{1}{2} x^2 e^{x/2}$$

$$y'' + 4y = 4 \sin(2x)$$

$$y'' + 4y \rightarrow m^2 + 4 = 0 \rightarrow m = \pm i2 \quad \bullet \text{ this is a case \#3}$$

$$y_c = c_1 \cos(2x) + c_2 \sin(2x)$$

$$y_p = A \cos(2x) + B \sin(2x)$$

A and c_1 ? Yes, so multiply by x

$$y_p = Ax \cos(2x) + B \sin(2x) \quad \leftarrow \text{use this going forward}$$

A and c_2 ? No repetition

B and c_1 ? No repetition

B and c_2 ? Yes, so multiply by x

$$y_p = Ax \cos(2x) + Bx \sin(2x) \quad \leftarrow \text{this will be your } y_p$$

$$y_p = Ax \cos(2x) + Bx \sin(2x)$$

$$\frac{Ax \cos(2x)}{f \quad g}$$

$$f' \cdot g + f \cdot g'$$

$$(A)(\cos(2x)) + (Ax)(-2 \sin(2x))$$

$$\frac{Bx \sin(2x)}{f \quad g}$$

$$f' \cdot g + f \cdot g'$$

$$(B)(\sin(2x)) + (Bx)(2 \cos(2x))$$

$$y'_p = A \cos(2x) - 2Ax \sin(2x) + B \sin(2x) + 2Bx \cos(2x)$$

$$y_p' = A \cos(2x) - 2Ax \sin(2x) + B \sin(2x) + 2Bx \cos(2x)$$

$$\frac{-2Ax \sin(2x)}{f \quad g}$$

$$f' \cdot g + f \cdot g'$$

$$\frac{2Bx \cos(2x)}{f \quad g}$$

$$f' \cdot g + f \cdot g'$$

$$(-2A)(\sin(2x)) + (-2Ax)(2\cos(2x))$$

$$(2B)(\cos(2x)) + (2Bx)(-2\sin(2x))$$

$$y_p'' = \underline{-2A \sin(2x) - 2A \sin(2x) - 4Ax \cos(2x)} + \underline{2B \cos(2x) + 2B \cos(2x) - 4Bx \sin(2x)}$$

$$y_p'' = -4A \sin(2x) - 4Ax \cos(2x) + 4B \cos(2x) - 4Bx \sin(2x)$$

equations:

$$y_p'' = -4A \sin(2x) - 4Ax \cos(2x) + 4B \cos(2x) - 4Bx \sin(2x)$$

$$y_p = Ax \cos(2x) + Bx \sin(2x)$$

$$(-4A \sin(2x) - 4Ax \cos(2x) + 4B \cos(2x) - 4Bx \sin(2x)) + 4(Ax \cos(2x) + Bx \sin(2x))$$

$$= 4 \sin(2x)$$

$$-4A \sin(2x) - 4Ax \cancel{\cos(2x)} + 4B \cos(2x) - 4Bx \cancel{\sin(2x)} + 4Ax \cancel{\cos(2x)} + 4Bx \cancel{\sin(2x)} = 4 \sin(2x)$$

$$-4A \sin(2x) + 4B \cos(2x) = 4 \sin(2x)$$

$$\cancel{-4A \sin(2x)} = \cancel{4 \sin(2x)}, \quad 4B \cos(2x) = 0$$

$$\boxed{A = -1}$$

$$\boxed{B = 0}$$

Write the general solution: $y = y_c + y_p$

$$y = C_1 \cos(2x) + C_2 \sin(2x) + Ax \cos(2x) + Bx \sin(2x)$$

$$y = C_1 \cos(2x) + C_2 \sin(2x) - x \cos(2x)$$

$$y'' - y' = -9$$

$$m^2 - m = 0 \rightarrow m(m-1) = 0 \rightarrow m=0, m=1 \quad \bullet \text{ Case \#1}$$

$$y_c = C_1 + C_2 e^x$$

$$y_p = A$$

A and C_1 ? Yes, there is repetition so multiply by x

$$y_p = Ax \leftarrow \text{use this one going forward}$$

A and C_2 ? No, repetition

$$y'_p = A$$

$$y''_p = 0$$

So, our y_p is: $y_p = Ax$

Plug y'_p and y''_p into the equation

$$(0) - (A) = -9$$

$$A = 9$$

Write the general equation: $y = y_c + y_p$

$$y = C_1 + C_2 e^x + 9A$$